



Investigation of Doppler Lidar for Velocity Measurements in Wind Tunnels

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Abstract

A low-cost single component lidar velocimetry technique was designed and tested using a 200-mW continuous wave (CW) laser at the Laser Optical Communication Institute (LOCI), University of Dayton to be used in the wind tunnels for non-intrusive velocity measurement. A numerical framework for a scaled down Lidar system was developed to predict the required laser power and detector sensitivity for multiple velocities and shorter target distances. The designed Lidar system was built and tested over a range of speeds (7.5 – 9 m/s) using a small-scale wind tunnel. A fast Fourier transform (FFT) was performed on the collected voltage data to determine the Doppler shift of reflected light. The Lidar system was used to measure one dimensional velocity in a 10 cm range between the contraction and the test section of a small-scale wind tunnel. This location was selected for the Lidar system to measure the flow acceleration. However, flow seeding was required to obtain superior SNR from the setup. The Lidar system was able to measure the range of velocities occurring within the 10cm range of the wind tunnel thereby identifying the net acceleration of the flow in the measurement region.

I. Nomenclature

A_{rec}	Aperture size of receiver optics, m^2	N_0	Number of photons transmitted by the laser
c	Speed of light, $3.00 \times 10^8 \frac{m}{s}$	P_{TX}	Transmitting power of the laser, W
E_{LO}	Electric field of the local oscillator, N/C	r	Distance to the focal point of the system, m
E_{sig}	Electric field of the Doppler shifted light, N/C	t	Integration time of the photodetector, sec
E_λ	Energy of a photon, J	V	Air velocity, m/s
f_{beat}	Beat frequency measured by the oscilloscope, Hz	V_{LOS}	Air velocity component along the beam, m/s
f_{laser}	Frequency of the local oscillator, Hz	w_0	Laser beam waist, m
f_{sig}	Frequency of the light returning through the collector lens, Hz	β_m	Rayleigh cross section per unit volume, $\frac{1}{nm^4}$
Δf	Doppler shift frequency, Hz	Γ	Probe length, m
h	Planck's constant, $6.626 \times 10^{-34} \frac{m^2 \cdot kg}{s}$	θ	Angle between the air velocity and the laser beam, deg.
I	Photodetector electrical output strength, A	λ	Laser wavelength, m
N	Number of photons returning to the photodetector	ω_{LO}	Frequency of the local oscillator, rad/sec
		ω_{sig}	Frequency of the Doppler shifted light, rad/sec

II. Acronyms

<i>AFRL</i>	Air Force Research Laboratory	<i>DAQ</i>	Data Acquisition
<i>APD</i>	Avalanche Photodetector	<i>FFT</i>	Fast Fourier Transform
<i>BD</i>	Beam Dump	<i>LDV</i>	Laser Doppler Velocimetry
<i>CWL</i>	Continuous Wave Laser	<i>LO</i>	Local Oscillator
<i>DAGSI</i>	Defense Associated Graduate Student Innovators	<i>LOCI</i>	Ladar Optical Communications Institute
		<i>PDF</i>	Probability Density Function

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PIV Particle Image Velocimetry
RF Reflective Filter
RPM Revolutions Per Minute
SNR Signal-to-Noise Ratio

UD-LSWT University of Dayton Low Speed
 Wind Tunnel
WPAFB Wright Patterson Air Force Base

III. Introduction

Particle Image Velocimetry (PIV) and Laser Doppler Velocimetry (LDV) are two of the most popular non-intrusive techniques used for velocity measurements in wind tunnels. However, both methods require seeding the tunnel using tracer particles which lead to uneven distribution of seeding, interference with instruments, and improper alignment with the flow during local acceleration or deceleration. Research is commencing at the University of Dayton Wind Tunnel (UD-LSWT) to test the use of heterodyne detection Doppler wind lidar (light detection and ranging, also referred to as ladar) as an alternative to PIV and LDV. The Doppler effect explains the relationship between velocity, V , and changes in electromagnetic wave frequency, Δf , via the following equation

$$\Delta f = \frac{2V}{\lambda} \quad (1)$$

where λ is the wavelength of the electromagnetic wave in question. If Δf is measured, the line of sight velocity along a laser beam can be calculated. In Eqn. 1, a positive and negative Doppler frequency shift represents air moving towards and away from the light source, respectively. The major difference between PIV/LDV and Doppler wind lidar is that the previous methods depend upon Mie scattering of light while lidar primarily depends upon Rayleigh scattering. In Mie scattering, the particles used for reflection have a diameter of comparable or larger size to the wavelength of light, such as the particles used for seeding wind tunnels. Rayleigh scattering involves light reflecting off particles much smaller than the wavelength of light, such as air molecules, that naturally exist in the atmosphere.

Wind lidar is currently used for many applications, one of which is measuring atmospheric mean and fluctuating velocity to study turbulence at altitudes of 2-50+ kilometers. [1] Commercial wind lidar systems are also sold to operate at shorter ranges of about 10-200+ meters. [2] These shorter-range systems are sometimes used to increase the safety of commercial flight by detecting wake vortex and wind shear hazards (such as microbursts) near airports in a timely manner [3]. When aircraft fly behind one another in close succession, the wingtip vortices pose a dangerous hazard. Doppler lidars are used for their detection which increases passenger safety and airport capacity. Project Green Wake1 [4] uses an Ultra-Violet (UV) lidar-based technique to detect the wingtip vortices and wind shear as well as anticipate and mitigate its effects on the aircraft and its occupants.

Another common application for the short-range wind lidar systems is measuring air velocities around wind turbines. [5] The flow around wind turbines can be highly unsteady and complex. Accurate estimation of wind speed is comparatively difficult in highly turbulent and/or complex terrain conditions, but having prior knowledge of airflow is important for turbine installations and to determine optimum turbine size, spacing, etc. One such area of study involving the use of lidar and wind turbines is wake-steering control. [6] Researchers want to study the wake behind a wind turbine in order to perform proximity analysis of the turbines and optimize the design and its operation. A schematic of operation using a commercially available steerable lidar for three-dimensional flow is shown in Figure 1. [7] In this scenario, three lidar systems work in unison to combine three line-of-sight measurements into one three-dimensional flow velocity at each point of measurement.

This same method of Doppler lidar can also be scaled down further for use in a wind tunnel at ranges of 1-2 meters. A modified ZephIR (now called ZX Lidars) 300 Lidar was used at the Denmark Technical University to make one-dimensional flow measurements in a wind tunnel with velocity results that compared well to pressure sensors and a Pitot tube. [8] In Italy, the WindScanner lidar system was adapted for use in a large wind tunnel where it could make measurements around scaled wind turbines. Working at their minimum range, a pair of WindScanners were able to make two-dimensional flow velocity measurements that matched well with hot wire anemometer measurements. [9] The normalized flow velocity in the downstream direction around three turbine models obtained from a pair of WindScanners is shown in Figure 2. [9]

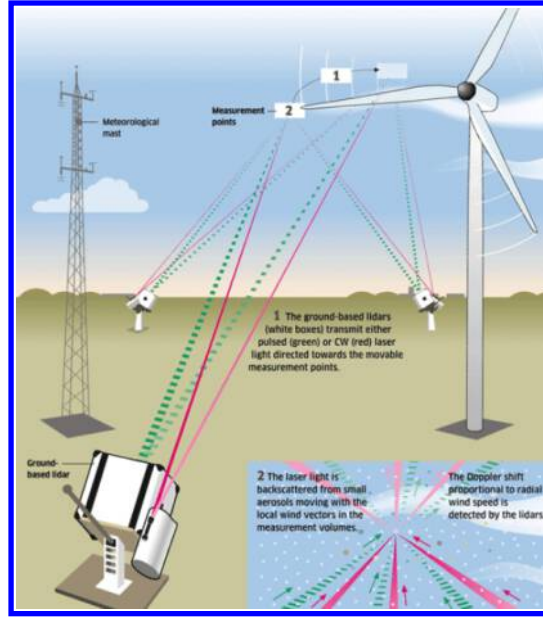


Figure 1. A visualization of a set of three lidar systems that scan in unison to create a 3D velocity profile. (Adapted from [7])

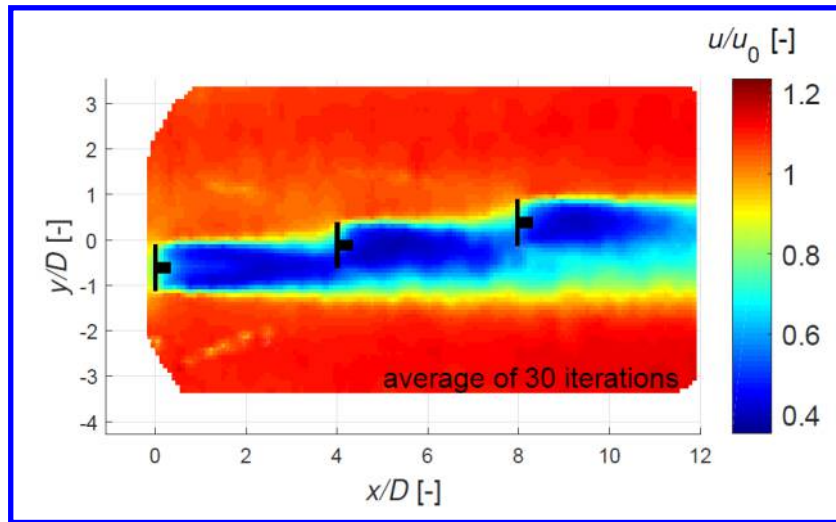


Figure 2. Normalized wind speed measurements are taken by two WindScanner lidar systems mounted outside of the flow in a wind tunnel. (Adapted from [9])

The biggest challenge with measuring air velocities using Doppler lidar is the minimum backscattering from air molecules. The return power in backscattering, P_R , when the object is smaller than the beam diameter, is proportional to

$$P_R \propto P_t A_0 \quad (2)$$

where P_t is laser transmitted power and A_0 is the object of interest's cross-sectional area. Seeding particles commonly utilized in wind tunnels have diameters in the range of 0.5 to 10 μm [10] while water vapor encountered in the atmosphere is approximately 2 to 12 μm in diameter. [11] On the other hand, air molecules have a diameter of $4 \times 10^{-4} \mu\text{m}$. [12] Using Eqn. 1, P_R reduces by a factor up to 625 million for a single particle/molecule. However, in a given volume, there are more air molecules present than tracer particles, so the return power is greater than the

relationship predicted using Equation 2. Figure 3 below shows the change in scattering intensity as the particle diameter reduces to sizes below common laser wavelengths of 532, 1064, and 1550 nm where Rayleigh scattering is observed rather than Mie scattering. [13] While the chart does not include the diameter of air molecules, it shows the extreme differences in the backscattering between air molecules and conventional tracer particles. Even though the goal of this research is to rely on scattering from air molecules rather than tracer particles to measure airspeed, it does pose fundamental technical challenges.

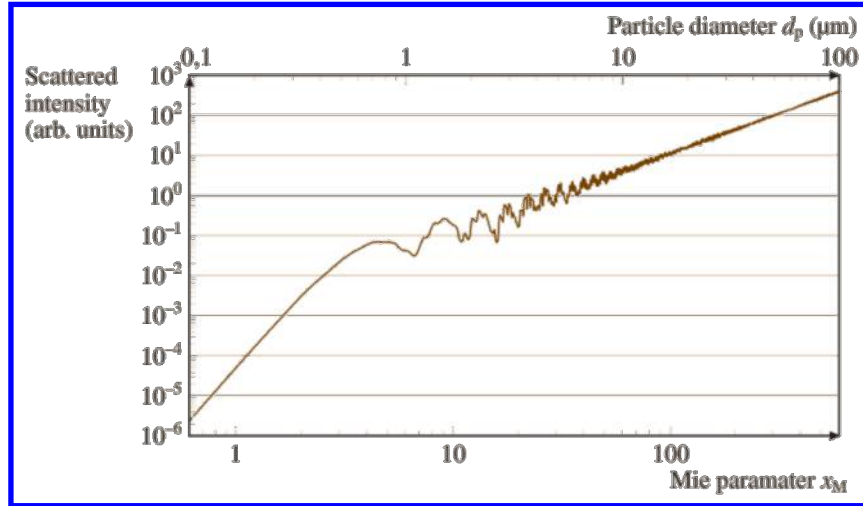


Figure 3. Calculated scattering intensity changes as a function of the Mie parameter and therefore particle diameter for a water droplet. (Adapted from [13])

IV. Experimental Setup

A. Beat Frequency Measurement

Determining air velocity requires a lidar setup that can measure the Doppler frequency shift of light waves that reflect off moving air molecules and aerosols. Figure 3 below depicts a general setup for heterodyne detection Doppler lidar detection in a wind tunnel. In a wind tunnel, the air is traveling at a velocity, V , which could be at an angle θ to the laser. The system measures velocity along the laser beam, V_{LOS} , where

$$V_{LOS} = V \cos(\theta) \quad (3)$$

The beam used to measure V_{LOS} exits the laser (CWL) and is sent through a beamsplitter, where most of the power is directed down range and through the region of interest before ultimately being collected by a beam dump (BD). Operating as a heterodyne detection system, a small amount of the beam is sent through a local oscillator (LO) where it is mixed with Doppler-shifted returning light at another beamsplitter before being projected onto a photodetector (APD) mounted across from the laser. The returning light is collected via a large plano-convex lens and collimated to maximize heterodyne mixing efficiency with the LO beam. This design is for a bistatic system, meaning that the outgoing laser beam does not pass through the collector lens. The lens must be angled slightly depending upon the measurement distance to collect the reflected photons from a desired location. Only light traveling directly at the beam with no incidence angle is collimated and reaches the APD through a final focusing lens, so the angle of this lens can be adjusted to select a measurement point of interest. A more robust way will include a pulsed laser with range gating techniques. At a range of 1.5 m, an approximately 10 cm section of the beam has reflected light that passes through the lens and reaches the APD. Therefore, any velocities within this 10 cm section are measured by the lidar system. If the velocity is consistent here, so is the Doppler shift. If the flow is accelerating, multiple velocities are measured due to varying Doppler shifts in different parts of the 10 cm collected area.

The APD is used to measure the frequency difference between f_{laser} and f_{sig} , called the beat frequency, f_{beat} ,

$$f_{beat} = |f_{sig} \pm f_{Laser}| = |f_{Laser} + \Delta f \pm f_{Laser}| = \Delta f \quad (4)$$

which has a low enough frequency to be measured with a photodetector [1]. Notice that f_{laser} and f_{sig} can also add to each other. The resulting frequency is well above the detection speeds of most equipment, so focus is only placed on f_{beat} where f_{laser} and f_{sig} subtract from each other. This f_{beat} is equivalent to the Doppler shift created by air molecules the light is reflected from. The LO consists of a nonpolarizing beamsplitter and a neutral density reflective filter (RF) used to reduce the power of the LO. A second beamsplitter is used to merge the LO and return signal beam before reaching the detector as a single beam.

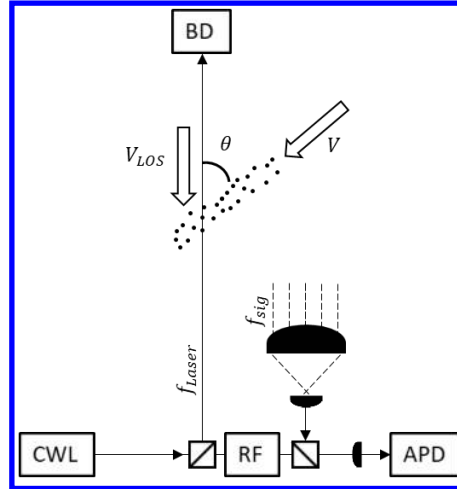


Figure 3. Basic components for heterodyne detection Doppler wind lidar detection

The f_{beat} received by the photodetector from illuminated aerosols and air molecules in addition to the LO can be read using an oscilloscope with high enough bandwidth. The post-processing of the signal involves reducing background noise and performing a fast Fourier transform (FFT) on the return signal to identify f_{beat} . Various sensitivity studies were performed to predict the changes in the required equipment specifications as a function of V . This analysis is discussed in a later section.

B. Laser Selection

The selection of a proper laser to be used in the lidar setup depends on multiple variables including safety, setup, range of wind tunnel speeds, and measurement distances. For safety reasons, a 532 nm wavelength laser was selected for the test setup. Since the intensity of Rayleigh scattering decreases with wavelength to the fourth power, shorter wavelengths such as 532 nm are usually preferred in comparison to 1064 nm or 1,550 nm. [1] Since Doppler shift is inversely proportional to the wavelength, shorter wavelength beams require high bandwidth equipment to detect f_{beat} .

A 532 nm Nd:YAG Quantel pulsed laser (200 mJ/pulse), used for PIV in the UD – LSWT, was considered as a plausible light source for the lidar setup. Given the velocity range of the UD – LSWT (1 – 37 m/s) and Eqn. 1, the Doppler shift range for these speeds is 3.7 – 136.9 MHz which is associated with wave time periods of 532 – 14 ns. A detector, to be discussed in detail later, needs to have a bandwidth twice as fast as the Doppler shift frequency for detection. The laser pulse width needs to be the length of the detector capture time (266 – 7 ns if perfectly matching twice the Doppler shift period) at an absolute minimum. However, the pulse width of the Quantel laser is around 4 ns, well below the pulse width needed to detect the tunnel speed. To detect a Doppler shift in the frequency, it is desired to have a laser pulse width greater than the expected bandwidth. Therefore, the Nd:YAG laser was not plausible as a light source for Lidar velocimetry.

If using a pulsed laser, the distance to the point of measurement and length of the LO needs to be the same so that the same pulse comes back together at the detector. Measuring f_{beat} of two separate pulses would result in inaccurate results because the wavelength can vary slightly between pulses. The roundtrip distance traveled by a laser beam in the UD-LSWT during testing is in the 2-6 m range which requires a laser coherence length of equal magnitude or greater to ensure that two different pulses are mixed. This distance was doubled to ensure that it would be long enough for any type of test setup and future use in a larger wind tunnel at the Air Force Research Laboratories (AFRL)

at Wright Patterson Air Force Base (WPAFB). These complications with using a pulsed laser led to selecting a continuous wave (CW) laser rather than a pulsed laser.

In addition to the other specifications described above, the necessary laser power was also determined. This selection was made based upon the number of transmitted photons reflected to the photodetector. The wavelength for a laser is used to calculate the energy in each photon, E_λ ,

$$E_\lambda = \frac{hc}{\lambda} \quad (5)$$

where h is Planck's constant and c is the speed of light. The use of laser power, P_{TX} , and photodetector integration time, t , resulted in a transmitted photon count of

$$N_0 = \frac{P_{TX}}{E_\lambda} t = \frac{P_{TX} \lambda}{hc} t \quad (6)$$

Other parameters driving the receiver photon count include the probe length and the parameter for the Rayleigh cross-section per unit volume in which the probe length, Γ , is

$$\Gamma = \frac{\lambda r^2}{\pi w_0^2} \quad (7)$$

In Eqn. 6, r is the distance to the focal point of the system and w_0 is the laser beam waist. The Rayleigh cross-section per unit volume, β_m , is defined as

$$\beta_m = 1.39 \left(\frac{550}{\lambda(nm)} \right)^4 \times 10^{-12} \quad (8)$$

where λ has units of nanometers. Equations 5-7 are used to create eqn. 8 which outputs the number of photons returning to the photodetector after being reflected off moving air molecules.

$$N = \frac{3}{16\pi\Gamma^2} N_0 c A_{rec} \beta_m \quad (9)$$

In Eqn. 8, A_{rec} is the aperture size for the receiving optics. When working inside wind tunnels, the atmospheric attenuation of light can be ignored due to the extremely short ranges. At higher wind tunnel speeds with a larger Δf and correspondingly lower N , fewer photons are reflected onto the APD. Figure 4 displays the variation in the number of returned photons as a function of laser power and range of air velocities. A 50mW laser at wind tunnel velocities upwards of 50 m/s that can be seen in wind tunnels at WPAFB returns only 185 photons integration time. This can barely be measured by linear mode avalanche photodetectors (which typically can measure a minimum of approximately 50-100 photons), and any changes in the experimental setup or losses would make the measurements impossible. Therefore, it can only really be measured by Geiger mode APDs that cost in the range of approximately \$5,000 compared to a Silicon linear mode APD which is ~25% of the price. Factoring in the cost of the laser and detector, a 100 or 200mW laser was preferred as the light source for lidar. After considering these calculations, a 532nm Cobolt Samba™ 200mW CW laser was selected to use for this project to maximize the photon return. [14]

C. Photodetector and Optics Selection

The photodetectors were also studied to determine which would be an appropriate match for the selected laser and anticipated wind tunnel speeds. Besides laser power selection, the greatest contributors to changes in N during testing are r and w_0 . Figure 5 shows how the number of returned photons changes with different ranges that could be seen in various lidar setups possible during testing at speeds up to 50 m/s.

Different collector lens diameters of 75 mm and 150 mm were to determine the changes in number of photos captured by the detector. Some of the major types of photodetectors include PIN diodes, linear mode APDs, and Geiger mode APDs with minimum photon count thresholds of approximately 400, 50, and 5, respectively, depending upon the detector chosen. With a 200mW laser, 75 mm receiver aperture, and a range of 1.5 m, the number of returning

photons falls below the PIN diode limit at speeds over 2.5 m/s, and the linear mode APD limit is reached at 18 m/s. Increasing the receiving lens diameter from 75 to 150 mm allowed the linear mode APD to be used in all expected velocities at a range of 1.5 m. Based on Figure 5, the system with the 150 mm collector lens would be limited to ranges below 1.7 m without requiring a Geiger mode APD. The only velocity measurement limitation is the DAQ bandwidth and sampling rate for capturing the signal.

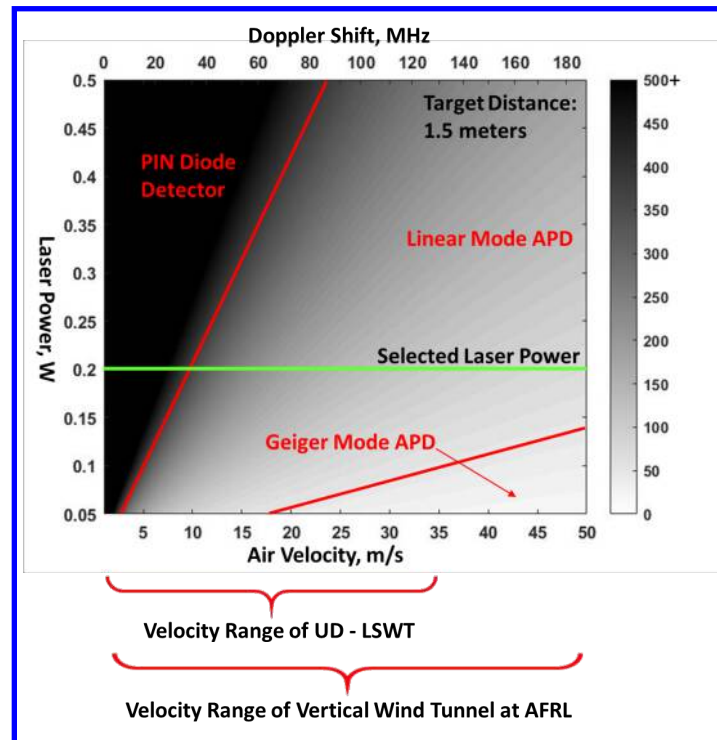


Figure 4. The change in returning photons that can be measured with different velocities and laser powers.

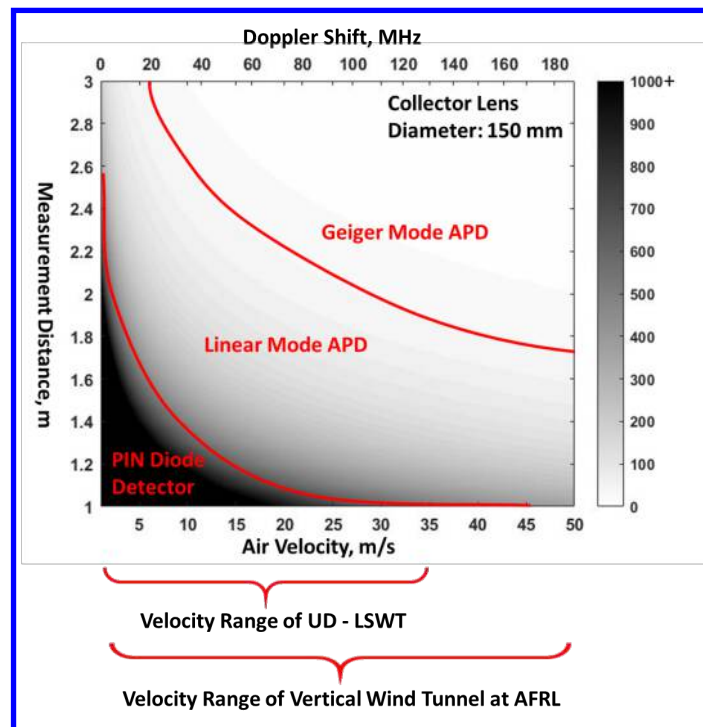


Figure 5. Change in photons returning to the photodetector with various possible measurement ranges at 50m/s

The third option for photon detection was a Geiger mode APD. The increased sensitivity of the Geiger mode APD, while beneficial for detecting weak signals, makes it susceptible to damage from a high level of photons reflected when operating at low speeds. In addition, Geiger mode APDs have a typical dead time of 400 ns to 1 μ s after detecting a single photon. A dead time of 1 μ s corresponds to a 2.5 MHz frame rate. A recently developed Geiger mode APD camera operates asynchronously with a 32x128 pixel format that could provide much higher frame rates. [15] A multiple pixel array helps achieve the desired speed, but the device is not practical to the experiment due to costs that are several orders of magnitude greater than other single-pixel detectors considered. Based on this information, a silicon, linear mode APD was selected to allow for a variety of wind tunnel speeds and ranges along with a large receiver lens. The selected linear mode APD (ThorLabs, APD430A2), has a bandwidth of 400 MHz which allows wind speed measurements greater than 50 m/s, above the speed limits of both wind tunnels in consideration. It is a UV compensated design for higher responsivity at the chosen wavelength of 532 nm since silicon APDs typically have the highest responsivity at wavelengths closer to 800 nm. In Figure 6, the selected laser wavelength is marked on the responsivity curve of several ThorLabs APDs. The blue line represents the APD430A2 which has a responsivity of approximately 46 A/W. [16] Additionally, a variable gain from 10 to 100 allows adjustments to be made based on the number of returning photons.

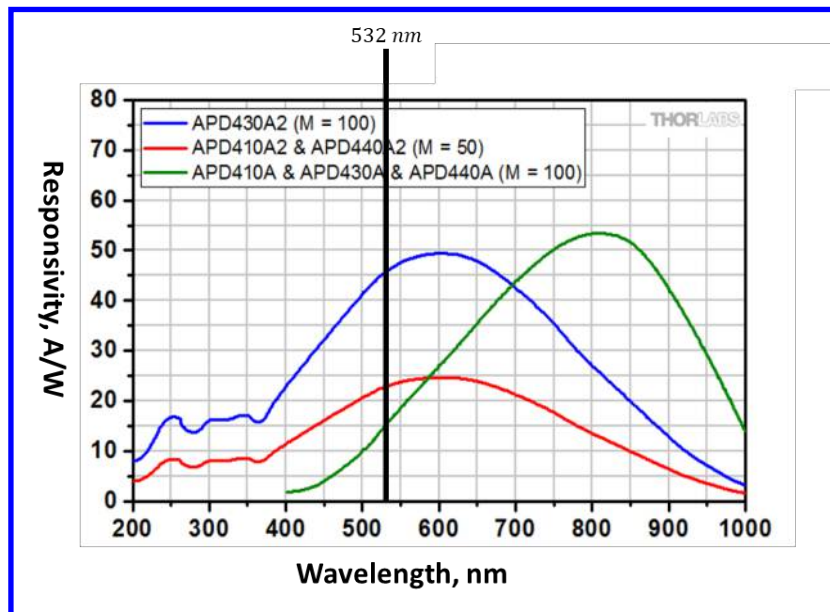


Figure 6. Responsivity for several ThorLabs photodetectors, including the selected APD. (adapted from [16])

The final primary component for the lidar setup is the LO which includes a 90:10 nonpolarizing beamsplitter cube and RF. A collimated beam exits the laser, 90% of which reflects off the beamsplitter and travels downrange into the moving airflow. The other 10% passing through the LO is still too powerful, so it passes through the RF, reducing the intensity to not oversaturate or damage the detector. Ultimately, less than 0.1% of the original beam exiting the laser is used in the LO as the reference beam (Figure 4). The second beamsplitter used to merge the beams is also a 90:10 splitter to maximize the amount of returned light reaching the APD.

The LO beam, while greatly reduced in intensity due to the beamsplitters and RF, must be stronger than the return signal. A larger LO helps to reduce the signal-to-noise ratio (SNR) of the analyzed data. [15] A strong LO causes shot noise, a type of white noise, to be the main component of noise. This type of noise is easier to filter out in post-processing. A major benefit of heterodyne detection lidar over direct detection lidar is that the return signal does not need to have a high intensity. As can be seen in Eqn. 9, the intensity of the electrical power output by the APD, I , is a function of both the return signal electric field, E_{sig} , and the LO electric field, E_{LO} . [15]

$$I = 2E_{sig}E_{LO}\exp[-j(\omega_{sig} - \omega_{LO})] \quad (10)$$

In Eqn. 9, ω_{sig} is the Doppler-shifted return signal frequency while ω_{LO} is the LO frequency. A strong APD output showing f_{beat} can be achieved with even a small E_{sig} when the LO intensity is increased. Within the LO, the RF is variable to allow different LO intensities depending upon the testing being conducted. This provided additional

flexibility to the system design while in preliminary stages. The selected laser emits a beam with 2.9 W/cm linear power density, well below the damage thresholds of both devices. To conclude, the LO components and devices described above were purchased to construct the setup shown in Figure 7 in preparation for testing. Most equipment in Figure 7 is mounted to sets of one-axis traverses for precision motion control. The linear mode APD is connected to an Agilent Technologies 100 MHz InfiniiVision DSO-X 3012A oscilloscope for initial measurements. To measure speeds greater than 13.5 m/s, a faster oscilloscope would be needed. This device is the foremost constraint in detection speed.

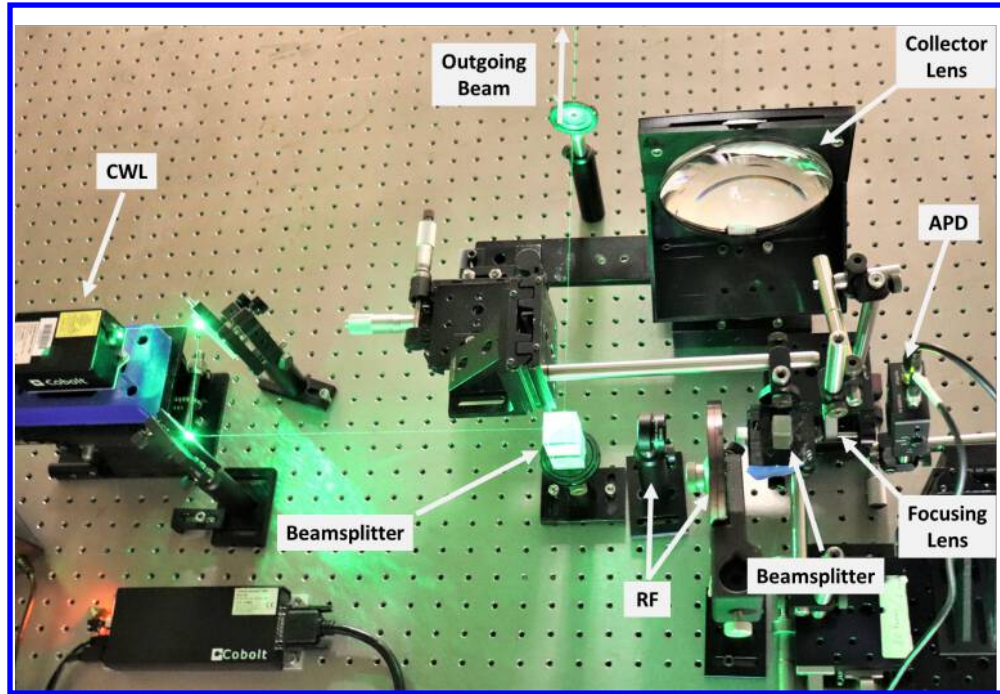


Figure 7. The constructed Doppler lidar setup as described using Figure 3.

V. Procedure

Initial testing was conducted to validate the test setup and that f_{beat} could be created by mixing the LO and return signal beams. The validation was first done by an ultrasonic piezoelectric transducer used to create high-speed vibrations of similar frequency to expected Doppler shifts. The transducer (5 MHz, STEMiNC, Part Number SMMSG20F5000), was set to operate at 1.44 MHz which allowed testing of the system as a vibrometer [17]. This device provides a vibration speed comparable in magnitude to what was expected with the Doppler shift. A high-speed vibration provides similar data to what can be expected when looking at the air in the next step. The validation of the test setup using the transducer provided a hard body for light to reflect off rather than relying on the reflection from air molecules for initial testing. Fine-tuning of the optical setup could then be made with this transducer operating to improve alignment and the measured frequency signal.

After refining the system optics to work optimally with the transducer, attention was turned to measure the Doppler shift from air molecules. A small-scale wind tunnel was used to create a steady flow at the location along the beam where the system focused. A Pitot tube was used to measure velocities inside of the wind tunnel for comparison to lidar measurements between points A and B, approximately 10 cm apart in the inlet as shown in Figure 8. At a low-speed setting, the velocity measurements at points A and B were 7.5 and 8.3 m/s, respectively. Increasing the speed of the wind tunnel slightly, new velocity measurements of 8.0 and 9.0 m/s were made. As mentioned above, the maximum velocity which could be estimated through the DAQ is 13.5 m/s. The lidar system was adjusted to measure within this same 10 cm section of the flow. Using a handheld steam generator, water vapor was added to the flow as a tracer to improve the return power. Using an estimated particle diameter of $10\ \mu\text{m}$, a water vapor has a stokes number of 0.090 for 9.0 m/s flow. Inserting these tracer particles gives the laser additional material to reflect off and therefore sends more photons to the APD for an improved signal, like the use of the transducer. Efforts are currently being undertaken to improve the Lidar setup to be used without any tracer particles.

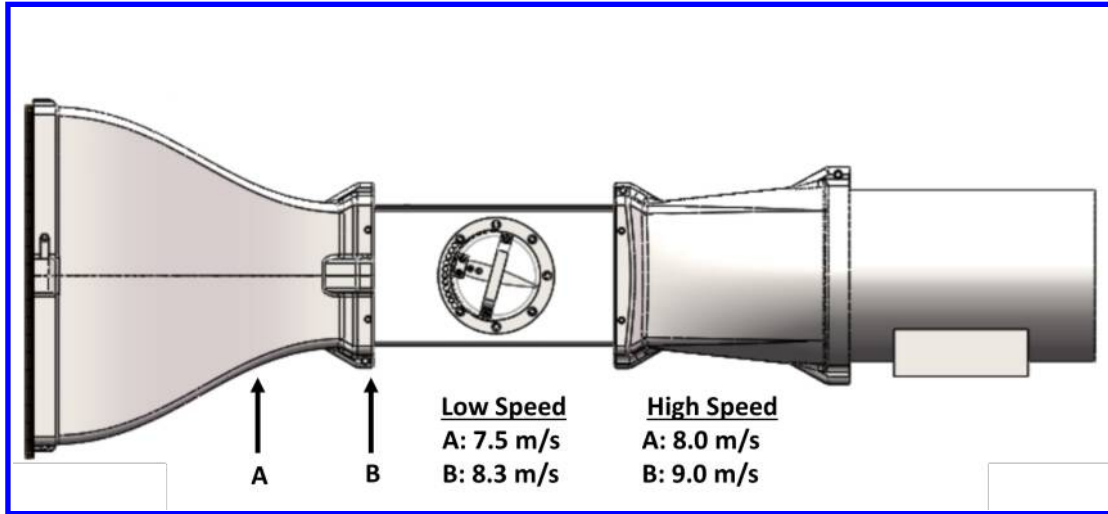


Figure 8. A small-scale wind tunnel was used to compare the Doppler lidar velocity measurements to a Pitot tube.

Primary method of data analysis consists of performing FFT on 200 μ s of voltage signal measured from the O-scope. Two hundred samples were collected and plotted together to assist in determining a range of dominant frequencies out of the noise in what is otherwise a weak signal. The data was recorded using an O-scope at a sampling rate at 80 MHz rather than its maximum 100 MHz to reduce noise while still being fast enough to capture the Doppler shift. Fourier transform of the data was performed which could detect frequencies up to 40 MHz, half the recording speed of the oscilloscope. The absolute magnitude of the signal in the frequency spectrum was raised to the sixth power after performing the FFT. This increased the SNR by a factor of 62.3. The Doppler shifts caused by the range of velocities between 7.5 and 8.3 m/s are 28.2 and 31.2 MHz, respectively. An amplitude peak between these frequencies was expected when performing an FFT of the raw data collected using the lidar setup. A weaker signal results in a lower signal to noise ratio making Doppler shift difficult to measure. A consistent signal peak would generate a peak that shifts minimally between data sets in contrast with noise that is random.

VI. Results

In Figure 9, the results of testing with the transducer in place are displayed. Multiple peaks of different amplitudes can be seen at 1.44 MHz, which is the frequency the transducer was operating. With a strong return signal due to the reflective metal surface, the return signal successfully mixed with the LO and created a measurable f_{beat} at the APD. This validation test showed that the system is properly aligned and working as a vibrometer.

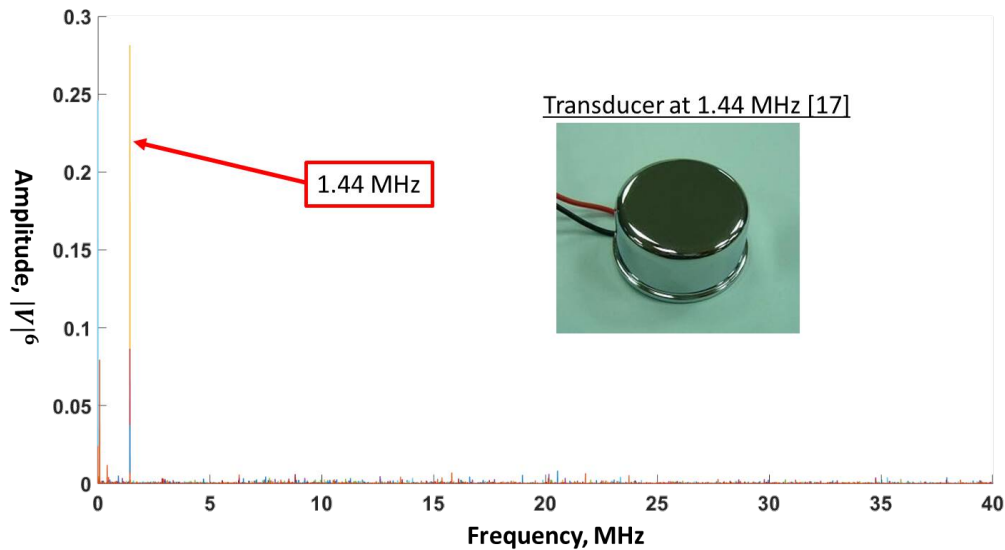


Figure 9. The lidar system can measure a transducer vibrating at a frequency of 1.44 MHz.

After validating the lidar setup with the transducer, lidar velocimetry was attempted using the small-scale wind tunnel shown in Figure 8 in the area where the velocity accelerates from 7.5 m/s to 8.3 m/s. The results are shown in Figure 10. A secondary x-axis at the top of the FFT plot shows the velocities associated with the measured Doppler shifts. To avoid noise in the analysis, only data with an amplitude greater than 100 in the frequency range of interest was studied. There is a clear high magnitude signal in the vicinity of 30 MHz. Out of the 200 data sets, a total of 28 peaks were found to have amplitudes greater than 100 within the velocity range of interest. Outside of this range, 435 peaks were found with an amplitude greater than 100, but they all were located at frequencies below 1 MHz and ignored because they were caused by the APD rather than Doppler shifts. The range of frequencies greater than 100 in amplitude ranges from 28.24 to 31.33 MHz which equate to 7.51 and 8.33 m/s, respectively as annotated in the figure which results in an acceleration of 61.5 m/s^2 measured in Eulerian frame of reference. These results match extremely well with the Pitot tube measurements. A 99.7% confidence interval of the frequency peaks (3 standard deviations from the mean frequency peak in each direction) includes frequencies between 27.59 and 32.44 MHz. This confidence interval is shown in Figure 10 below. All frequency peaks are within this range, so no data sets are outliers. It is important to note that each peak between the dotted lines shown in Figure 10 represents the measured velocity along the length of the light beam (10 cm) in focus. However, using the current setup, exact velocity measurement at one single point is not possible due to the incapability of the setup to focus on single point in space.

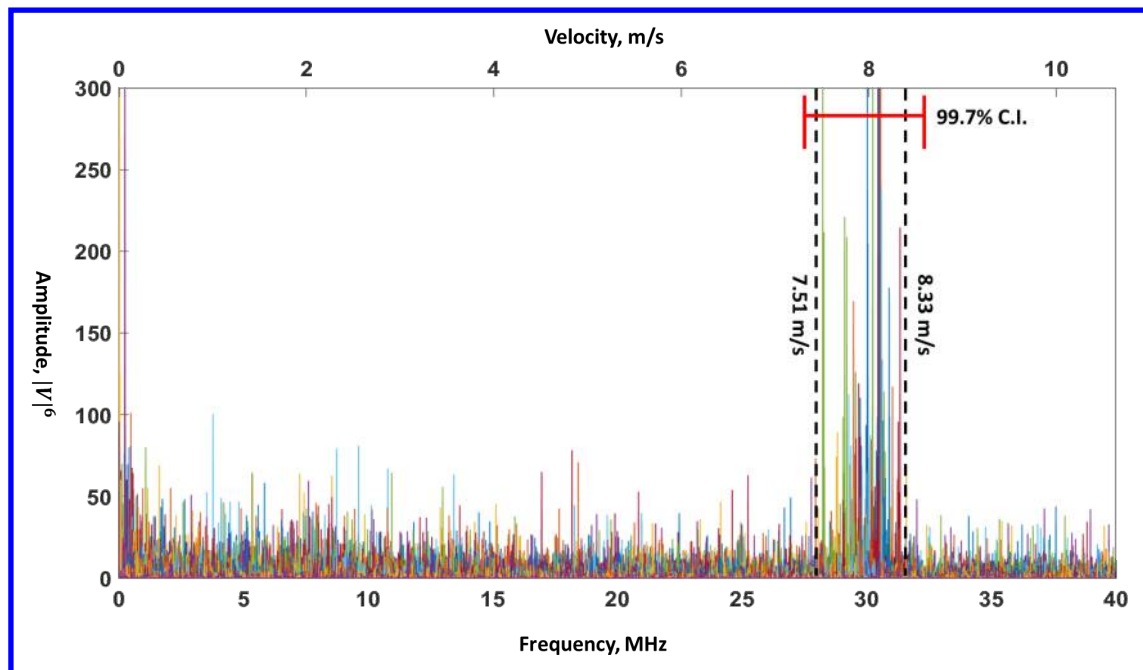


Figure 10. At the low-speed setting, a Pitot tube measured velocities between 7.5 and 8.3 m/s.

Given the successful agreement between Pitot tube and Lidar system results at lower velocities, the RPM of the small-scale wind tunnel was increased where the tunnel centerline velocity accelerated from 8.0 to 9.0 m/s between points A and B. The results from the Lidar setup are shown in Figure 11. High amplitude peaks (greater than $100 V^6$) can be seen between 30.42 to 33.78 MHz which corresponds to velocity of 8.09 to 8.98 m/s, respectively which results in an acceleration of 72 m/s^2 measured in Eulerian frame of reference. Again, the lidar and Pitot tube measurements match extremely well. One additional peak in the area of interest is at 36.30 MHz which corresponds to 9.65 m/s. A 99.7% confidence interval of the frequency peaks includes frequencies between 29.99 and 33.67 MHz. This peak is well outside of that range and considered an outlier. The major concentration of velocity measurements are from within the 10 cm measurement area.

It can be seen in Figure 10 and Figure 11 that some peaks within the measurement range are stronger than others. This may be due to the non-uniformity in water vapor seeding during the data acquisition resulting in non-uniform backscattering and mixing with the LO. Most measurements plotted have a peak at only one frequency or a small range of frequencies. The flow along the beam within the 10 cm measurement area is not always seeded evenly throughout, so when a measurement is taken, reflected light may be only strong enough to generate a visible signal in certain parts of the beam length being studied, with varying signal strength. One data set from the high-speed testing

is responsible for 18 of the 26 frequency peaks. Between the low and high velocity tests, 3% and 8.5% of the collected data sets contain the detected velocity measurements. At higher speeds, the water vapor emitted from the steam generator spread out more and were not as concentrated when they traveled through the measurement area. This caused a lower return to the detector. Despite this, there were still 26 total peaks with an amplitude greater than 100 which is only two less than that of the low-speed case. The total number of peaks is concentrated into less data sets.

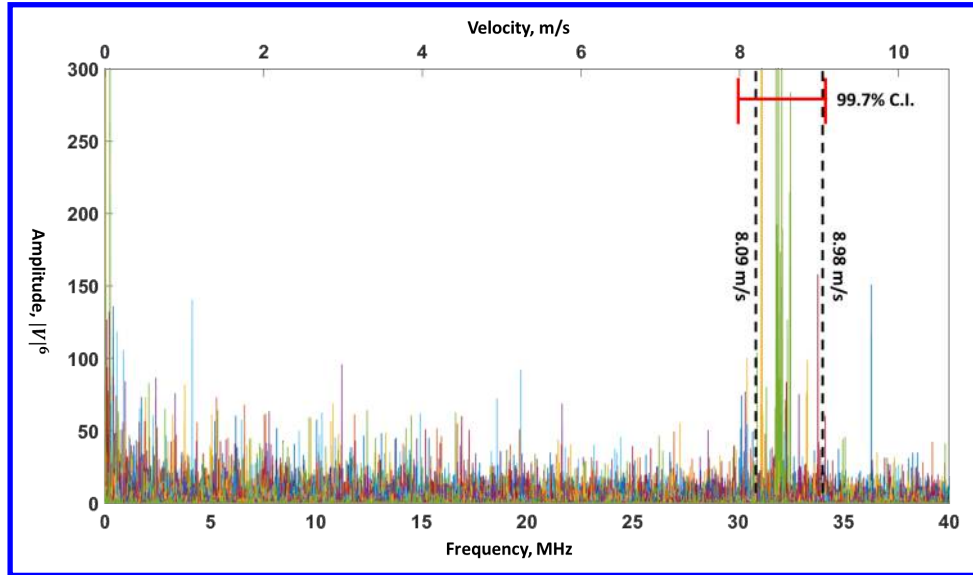


Figure 11. At the high-speed setting, a Pitot tube measured velocities between 8.3 and 9.3 m/s.

When these tests were repeated without seeding the flow with water vapor, no significant peaks were detected in the FFT despite the predictions by the numerical framework mentioned in the sections above. Through calculations, it was estimated that there would be 440 photons per detection returned to the APD at a range of 1.5 m and speed of 8.3 m/s. These calculations used for sizing the laser and selecting the APD did not account for various losses in the system which resulted in not enough photons returning to the APD. First, the optical efficiency of the system reduces the number of photons that successfully travel from the first collector lens to the 0.2 mm diameter photodiode in the APD. It is possible that some photons returning from the air molecules are reflected as they go through the return leg of the optical train, especially when combining with the LO at the beamsplitter. Despite having antireflective coatings, light can be seen reflecting off glass at every medium change throughout the optical train. Due to the size of the pixel in the linear mode APD detector, it is also possible that some of the photos could not be captured. Second, the heterodyne mixing efficiency was and still is unknown. This efficiency is directly related to the intensity of the signal that contributes to a good SNR. Third, the required SNR for analysis and detection of clear peaks in the FFT was unknown and unaccounted for.

VII. Conclusions

Proof of concept testing of lidar based velocimetry technique using a low powered laser was accomplished at the University of Dayton LOCI. A numerical framework to determine the appropriate laser power, photodetector, and LO for a new heterodyne detection Doppler lidar velocimetry technique for use at the UD-LSWT and in the wind tunnels at AFRL was developed. A strong signal to noise ratio was achieved by seeding the flow and raising the amplitudes of the FFT to the sixth power. Initial tests of water vapor seeded flow showed that one-dimensional lidar velocity measurements matched well with Pitot tube measurements inside of a small-scale wind tunnel. The range of velocities in a 10-cm measurement region was captured accurately to estimate the spatial acceleration of the flow. Only a few percentage of the entire data set contained good SNR in the frequency range of interest, indicating the need for higher number of samples.

VIII. Acknowledgments

The authors sincerely thank the Defense Associated Graduate Student Innovators (DAGSI) for funding the research presented. Recognition also goes out to the University of Dayton Ladar Optical Communications Institute (LOCI), specifically Cullen Bradley and Dr. Paul McManamon, for allowing me to use their optics lab and giving advice about improving the test setup. Finally, the assistance of AFRL sponsor Benjamin Hagen was extremely appreciated, providing critical equipment for testing.

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